CURRICULUM VITAE

NAME: Dr. Rebeccah Moraa Ayako

PROFESSIONAL PROFILE

An immunologist and research scientist with extensive experience in infectious diseases, immunotherapy, and molecular biology techniques. Skilled in PBMC isolation, ELISA, ELISPOT, flow cytometry, PCR, sequencing, bioinformatics, and tissue culture. Demonstrated expertise in malaria, leishmaniasis, HIV, and SARS-CoV-2 research using both in vitro and in vivo models. Passionate about advancing immunotherapy strategies for neglected tropical diseases and contributing to vaccine and drug development.

WORKING HISTORY

- **Kenya Institute of Primate Research (2025-present):** Research Scientist, Tropical and Infectious Diseases
- **Linkoping University (2023):** Visiting Researcher, Division of Molecular Medicine and Virology
- **Moi University (2021-2025):** Tutorial Fellow, College of Health Sciences: Volunteer
- **AMPATH, Kenya (2021- to 2022):** Volunteer: Research and diagnosis of Leukemia and Lymphoma:

EDUCATION AND TRAINING

11/2021 to 12/25: Doctor of Philosophy in Immunology, Moi University, Kenya

- ***Thesis:*** Immune responses in SARS-CoV-2 patients at Moi Teaching and Referral Hospital, Eldoret Kenya
- ***Supervisors:*** Prof. Kirtika Patel, Dr. Isaac Ndede and Prof. Marie Larsson

01/2018 to 12/2021: Master of Science in Immunology, Kenyatta University, Kenya

- ***Thesis***: Characterization of concordant visceral leishmaniasis and malaria infections in BALB/c mice.
- ***Supervisors***: Dr. Joshua Mutiso and Dr. Lucy Ochola

09/2013 - 05/2017: Bachelor of Science in Biology, Kenyatta University, Kenya

- ***Grade***: Second Class Honors (Upper Division)
- ***Research Project***: Antischistosomal effects of *Carica papaya* crude extracts on cercariae; an infective stage of *Schistosoma mansoni*.
- ***Supervisors***: Dr. Martin Mutuku and Mr. Sammy Orinda.

05/2014- 09/ 2014: Pursued Web Designing at Tracom College, Kenya.
Techniques learned: MySQL, PHP, JavaScript

CONTINUOUS PROFESSIONAL DEVELOPMENT

PRESENTATIONS (SELECTED)

1. **May 2026**: Kenya National Public Health Initiative, Mombasa, Kenya: **Poster**
2. **October 2025**: 5th Africa International Biotechnology & Biomedical Workshop, Nairobi, Kenya: **Poster Presentation and Talk**
3. **October 2023:** Medical Inflammation and Infection Center (MIIC), Rimforsa, Sweden: **Poster Presentation and Talk**

WORKSHOPS (SELECTED)

1. **November 2025**: KIPRE 20th International Conference, Nairobi, Kenya.
2. **October** 2025: Laboratory Animal Science (LAS) Education, Nairobi, Kenya.
3. **February 2023:** Grant and Manuscript writing, Mount Kenya University, Kenya

PUBLICATIONS AND AWARDS

Publications

1. **Ayako, R. M**., Patel, K., Ndede, I., Mining, S. K., Klingström, J., Nordgren, J., & Larsson, M. **(2026)**. Low

Systemic IFN Response and High Viral Load Are Associated with COVID-19 Disease Severity in Unvaccinated Patients in Kenya, 2022–2023. *COVID*, *6*(3), 51. https://doi.org/10.3390/covid6030051

2. Njeri, Judith, Joshua M. Mutiso, Pullary M. Huma, **Rebeccah M. Ayako**, and Michael M. Gicheru. **(2026)**. Synergistic Immunostimulation of *Carissa Edulis* Extract with *Leishmania donovani* Antigen in BALB C mice. South Asian Journal of Parasitology 9 (1):182-96. https://doi.org/10.9734/sajp/2026/v9i1265.
3. **Ayako, R. M**., Patel, K., Ndede, I., Mining, S. K., Klingström, J., Nordgren, J., & Larsson, M. **(2025)**. Low systemic IFN response and high viral load are associated with COVID-19 disease severity in unvaccinated patients in Kenya, 2022-2023. *Authorea Preprints*. https://doi.org/10.22541/AU.174768064.40021112/V1
4. Nyachieo, R.N., Maloba, F., **Ayako, R.M**. and Ngalla, J. **(2025)**. Exacerbated Hepatic, Splenic, and Neuropathology in BALB/c Mice Co-Infected with *Leishmania major* and *Plasmodium berghei*. Open Journal of Pathology, 15, 196-215. doi: 10.4236/ojpathology.2025.154016
5. **Ayako, R.M**., Patel, K., Ndede, I., Nordgren, J., Larrson, M. and Mining, S.K. **(2024**), Inflammatory, Hematological, and Biochemical Biomarkers in COVID-19 Patients. Immunity, Inflammation and Disease, 12: e70078. https://doi.org/10.1002/iid3.70078
6. Simbauni J, Kavuso M, Mutiso J, **Ayako R**, & Gicheru M. (2024). Foot and Mouth Disease Virus Strain-specific Antibody Titres in Naturally Infected or Vaccinated Bulls in Kenya. Asian Journal of Advances in Agricultural Research. **2024**. 1-9. 10.9734/ajaar/2024/v24i8531. https://www.researchgate.net/publication/382482549_Foot_and_Mouth_Disease_Virus_Strain-specific_Antibody_Titres_in_Naturally_Infected_or_Vaccinated_Bulls_in_Kenya
7. **Ayako Rebeccah**. M., Mutiso Joshua. M., Macharia John. C., Langoi D., & Ochola, L. (**2021**). Concomitant infection with *Leishmania donovani* and *Plasmodium berghei* causes Pro-inflammatory polarization resulting in malaria exacerbation in BALB/c mice. *American Journal of Infectious Diseases and Microbiology* http://pubs.sciepub.com/ajidm/9/3/1/index.html
8. Mainye, N., Maranga, D. N., **Ayako, R.,** & Ochola, L. A. (**2022**). *Machine learning meets microscopy: cell explorer tool for a diagnostic laboratory*. https://www.scienceopen.com/document?vid=3f768d69-f44c-4de6-a49e-9f74efeae26f
9. Maria Divinah Mogaka, Joshua M. Mutiso, John C. Macharia, **Rebeccah M. Ayako**, Bernard Osero, Michael M. Gicheru. (**2022**). Evaluation of therapeutic potential of *Warbugia ugandensis, Prunus Africana* and *Piliostigma thonningii* against *Leishmania donovani* in vitro and in BALB/c mice.

Journal of Zoological Research.

https://journals.bilpubgroup.com/index.php/index/search?query=Prunus

Awards

- **2025**: KIPRE 20th international Conference, Nairobi, Kenya.
- **2023:** Medical Inflammation and Infection Center (MIIC), Rimforsa, Sweden
- **2021:** 6th Africa International Biotechnology & Biomedical Workshop, Kisumu, Kenya
- **2021:** 11th African Congress of Immunology
- **2021:** AAI Introductory Immunology Course 2021
- **2021:** Woods Hole Immuno-parasitology
- **2021:** Beyond Science Initiative

LAST DATE PROMOTED

None

REFERENCES

Dr. Atunga Nyachieo	***Prof. Simeon Mining***	***Dr. Joshua Mutiso***
Director of Research and Product Development	*Professor of Immunology*	*Senior Lecturer*
Tropical and Infectious Diseases	*School of Medicine*	*Department of Zoology*
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Signature:

Date: 29th April 2026